

SDS 4030 - Statistics for Data Science II

Final Project Code

Problem 1. *Proof.*

```
# loading dataset Here
```

```
horse = read.table("horse.txt", header = TRUE)
head(horse)
```

```
##  surgery   age temperature pulse respiratory_rate temperature_of_extremities
## 1      0 adult      38.5     66                28                      cool
## 2      0 adult      38.3     40                24                      normal
## 3      1 young      39.1    164                84                      cold
## 4      1 adult      37.9     48                16                      normal
## 5      0 adult      38.6     80                36                      cool
## 6      1 adult      38.1     66                12                      cool
##  peripheral_pulse      pain peristalsis abdominal_distension
## 1      reduced continuous      absent                      severe
## 2      normal      mild hypomotile                      none
## 3      normal depressed      absent                      severe
## 4      normal      mild hypomotile                      moderate
## 5      absent      severe      absent                      severe
## 6      reduced      mild hypomotile                      none
##  packed_cell_volume total_protein      outcome surgical_lesion
## 1              45          8.4      died          0
## 2              33          6.7     lived          0
## 3              48          7.2      died          1
## 4              37          7.0     lived          1
## 5              38          6.2 euthanized          1
## 6              44          6.0     lived          1
```

```
# starter EDA on resp rate
```

```
library(ggplot2)
```

```
resp_summary <- summary(horse$respiratory_rate)
resp_summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.00   17.75   28.00   29.96   36.00   90.00
```

```
mean_resp <- mean(horse$respiratory_rate, na.rm = TRUE)
median_resp <- median(horse$respiratory_rate, na.rm = TRUE)
min_resp <- min(horse$respiratory_rate, na.rm = TRUE)
max_resp <- max(horse$respiratory_rate, na.rm = TRUE)

cat("Mean:", mean_resp, "\n")
```

```
## Mean: 29.95833
```

```
cat("Median:", median_resp, "\n")
```

```
## Median: 28
```

```
cat("Min:", min_resp, "\n")
```

```
## Min: 2
```

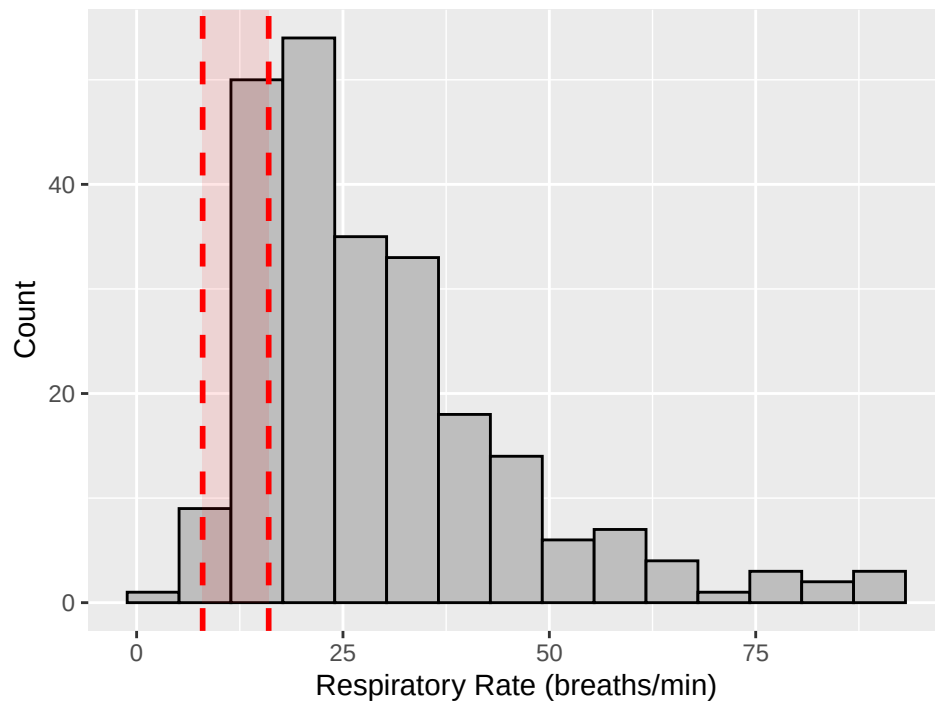
```
cat("Max:", max_resp, "\n")
```

```
## Max: 90
```

```
ggplot(horse, aes(x = respiratory_rate)) + geom_histogram(bins = 15,
  fill = "gray", color = "black") + geom_vline(xintercept = 8, linetype = "dashed",
  color = "red", size = 1) + geom_vline(xintercept = 16, linetype = "dashed",
  color = "red", size = 1) + annotate("rect", xmin = 8, xmax = 16,
  ymin = 0, ymax = Inf, alpha = 0.1, fill = "red") + labs(title = "Respiratory Rate Di",
  x = "Respiratory Rate (breaths/min)", y = "Count")
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Respiratory Rate Distribution with Normal Range (8–16)



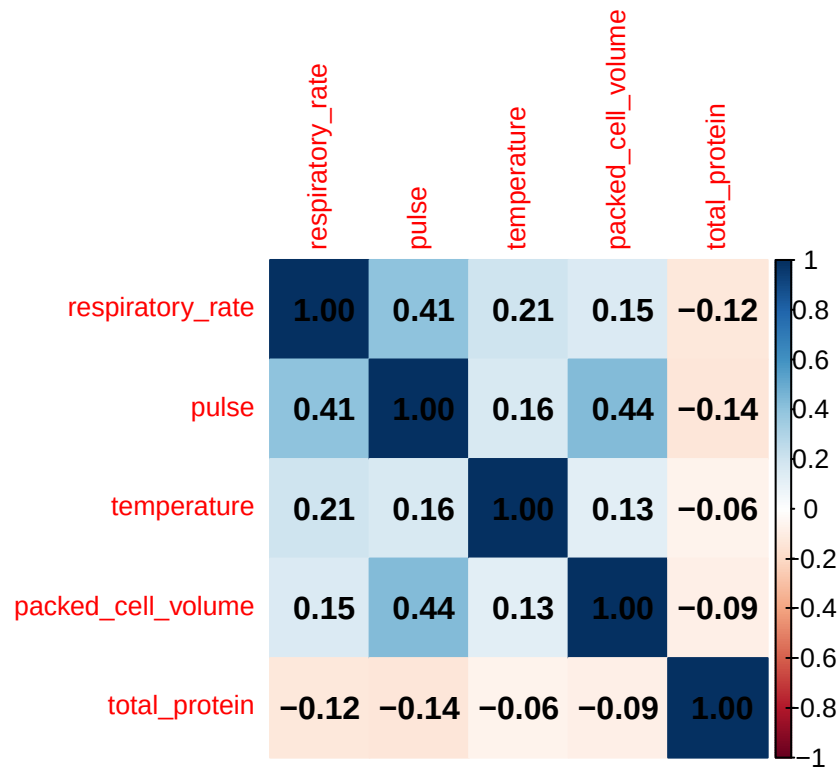
```
# correlation matrix for now
```

```
library(corrplot)
```

```
numeric_vars <- c("respiratory_rate", "pulse", "temperature", "packed_cell_volume",  
  "total_protein")
```

```
corr_matrix <- cor(horse[, numeric_vars], use = "complete.obs")
```

```
corrplot(corr_matrix, method = "color", addCoef.col = "black", tl.cex = 0.8)
```



```
# fitted model w only quant variables
```

```
quant_model = lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
  total_protein, data = horse)
summary(quant_model)
```

```
##
## Call:
## lm(formula = respiratory_rate ~ temperature + pulse + packed_cell_volume +
##     total_protein, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -32.852 -10.009  -3.102   7.040  56.187
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -112.98010   52.66873  -2.145   0.0330 *
## temperature     3.39913    1.39221   2.442   0.0154 *
## pulse          0.25250    0.04231   5.967 8.82e-09 ***
## packed_cell_volume -0.06932    0.10832  -0.640   0.5228
## total_protein  -0.03715    0.03676  -1.011   0.3132
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.32 on 235 degrees of freedom
## Multiple R-squared:  0.1916, Adjusted R-squared:  0.1779
## F-statistic: 13.93 on 4 and 235 DF,  p-value: 3.282e-10
```

```
# qual, exact, large sample size -- lecture 21 reference
```

```
beta <- coef(quant_model)
X <- model.matrix(quant_model)
Y = horse$respiratory_rate
n <- nrow(X)
p <- ncol(X)
S2 <- sum(residuals(quant_model)^2)/(n - p)

R <- matrix(c(0, 5, 0, 0, 350, 0, 0, 40, 100, 0), nrow = 2, byrow = TRUE)

r <- c(3, 3)
k <- length(r)

FT <- crossprod(R %*% beta - r, solve(R %*% solve(t(X) %*% X) %*% t(R),
  R %*% beta - r))/(k * S2)
print(FT)
```

```
##           [,1]
## [1,] 0.002356961
```

```
pvalue <- pf(FT, k, n - p, lower.tail = FALSE)
print(pvalue)
```

```
##           [,1]
## [1,] 0.9976458
```

```
quant_reduced <- lm(respiratory_rate ~ I(temperature + total_protein +
  pulse + packed_cell_volume), data = horse)
summary(quant_reduced)
```

```
##
## Call:
## lm(formula = respiratory_rate ~ I(temperature + total_protein +
```

```
##      pulse + packed_cell_volume), data = horse)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -28.046 -12.065  -3.351   6.948  62.740
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                      12.21057     4.89454
## I(temperature + total_protein + pulse + packed_cell_volume) 0.10100     0.02719
##                                     t value Pr(>|t|)
## (Intercept)                      2.495 0.013284
## I(temperature + total_protein + pulse + packed_cell_volume) 3.715 0.000254
##
## (Intercept)                      *
## I(temperature + total_protein + pulse + packed_cell_volume) ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.47 on 238 degrees of freedom
## Multiple R-squared:  0.0548, Adjusted R-squared:  0.05083
## F-statistic: 13.8 on 1 and 238 DF, p-value: 0.0002536
```

```
anova(quant_reduced, quant_model)
```

```
## Analysis of Variance Table
##
## Model 1: respiratory_rate ~ I(temperature + total_protein + pulse + packed_cell_volum
## Model 2: respiratory_rate ~ temperature + pulse + packed_cell_volume +
##      total_protein
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      238 64523
## 2      235 55183  3    9339.9 13.258 5.016e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# qual, asymptotic, large sample size -- lecture 21 reference
```

```
WT = k * FT
print(WT)
```

```
##           [,1]
## [1,] 0.004713923
```

```
pvalue = pchisq(WT, k, lower.tail = FALSE)
print(pvalue)
```

```
##           [,1]
## [1,] 0.9976458
```

```
anova(quant_reduced, quant_model, test = "Chisq")
```

```
## Analysis of Variance Table
##
## Model 1: respiratory_rate ~ I(temperature + total_protein + pulse + packed_cell_volum
## Model 2: respiratory_rate ~ temperature + pulse + packed_cell_volume +
##      total_protein
##   Res.Df    RSS Df Sum of Sq  Pr(>Chi)
## 1      238 64523
## 2      235 55183   3    9339.9 1.189e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The conclusions agree of both the exact and asymptotic tests agree with each other. Both had a p-value of 0.9043546, so we fail to reject the null hypothesis. The Wald statistic and Exact F statistic are related as

$$W_n = \frac{\sigma}{S_{nQ1}^2} = kF$$

where F is the F statistic from the exact test. So we would expect the conclusions of both tests to agree with each other.

□

Problem 2. *Proof.*

```
# jackknife
betajack <- matrix(0, n, p)
for (i in 1:n) {
  jack <- lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
    total_protein, data = horse[-i, ])
  betajack[i, ] <- coef(jack)
}
betabar <- colMeans(betajack)
print(betabar)
```

```
## [1] -112.97355280    3.39896234    0.25248853   -0.06931150   -0.03715342
```

```
SE <- apply(betajack, 2, sd) * sqrt((n - 1)^2/n)
print(SE)
```

```
## [1] 57.37518872  1.53850267  0.04898266  0.12188824  0.03813573
```

```
# nonparametric bootstrap
nboot <- 10000
betaboot <- matrix(0, nboot, p)
set.seed(123)

for (k in 1:nboot) {
  indices <- sample(1:n, n, replace = TRUE)
  boot <- lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
    total_protein, data = horse[indices, ])
  betaboot[k, ] <- coef(boot)
}

betabar <- colMeans(betaboot)
print(betabar)
```

```
## [1] -110.53315091    3.33575200    0.25106308   -0.06812419   -0.03683839
```

```
SE <- apply(betaboot, 2, sd)
print(SE)
```

```
## [1] 57.58995133  1.54368567  0.04753332  0.11860789  0.03728970
```

```
alpha <- 0.05

CInormal <- cbind(betabar - qnorm(1 - alpha/2) * SE, betabar + qnorm(1 -
  alpha/2) * SE)
print(CInormal)
```

```
##           [,1]      [,2]
## [1,] -223.4073814 2.34107957
## [2,]   0.3101837 6.36132031
## [3,]   0.1578995 0.34422667
## [4,]  -0.3005914 0.16434300
## [5,]  -0.1099249 0.03624808
```



```
CIpercentile <- cbind(apply(betaboot, 2, quantile, probs = alpha/2),
  apply(betaboot, 2, quantile, probs = 1 - alpha/2))
print(CIpercentile)
```

```
##           [,1]      [,2]
## [1,] -221.4820617  5.61483854
## [2,]   0.2249939  6.28848895
## [3,]   0.1545856  0.33876000
## [4,]  -0.2986055  0.16237066
## [5,]  -0.1089141  0.03646585
```

```
beta_full <- coef(quant_model)
CIpivotal <- cbind(2 * beta_full - apply(betaboot, 2, quantile, probs = 1 -
  alpha/2), 2 * beta_full - apply(betaboot, 2, quantile, probs = alpha/2))
print(CIpivotal)
```

```
##           [,1]      [,2]
## (Intercept) -231.5750365 -4.47813629
## temperature   0.5097786  6.57327363
## pulse         0.1662342  0.35040860
## packed_cell_volume -0.3010173  0.15995882
## total_protein -0.1107758  0.03460421
```

```
summary(quant_model)$coefficients[, "Std. Error"]
```

```
##           (Intercept)           temperature           pulse packed_cell_volume
##           52.66872917           1.39221025           0.04231417           0.10832383
##           total_protein
##           0.03675915
```

The coefficient estimates are essentially the same across methods (OLS / jackknife mean / bootstrap mean). Both resampling methods give very different estimates than OLS. The Jackknife SEs and bootstrap SEs are very similar. These SEs are much larger than OLS SEs. The two sets of intervals are very similar to one another, further supporting the stability of the bootstrap results. Compared to OLS confidence intervals, the bootstrap intervals are significantly wider.

□

Problem 3. *Proof.*

```
# Nonparametric test of staistical significant: Permutation F Test
```

```
f = summary(quant_model)$fstatistic[1]
print(f)
```

```
##      value
## 13.92634
```

```
pvalue = pf(f, p, n - p - 1, lower.tail = FALSE)
print(pvalue)
```

```
##      value
## 6.431426e-12
```

```
nperm = 10000
fperm = numeric(nperm)
for (k in 1:nperm) {
  Yperm = sample(Y)
  perm = lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
    total_protein, horse)
  fperm[k] = summary(perm)$fstatistic[1]
}
pvalue = (1 + sum(fperm > f))/(1 + nperm)
print(pvalue)
```

```
## [1] 9.999e-05
```

The estimated F test p-value through the permutation method is equal to $1e-4$. Since the pvalue is less than alpha, we can reject the null hypothesis. The pvalue under the theoretical null distribution is $6.4e-12$, which is smaller than the pvalue from the permutation test. However, they both lead to the rejection of the null.

```
# Nonparametric test of statistical significance for the effect of
# temperature Permutation t Test
```

```
t = summary(quant_model)$coefficients["temperature", 3]
print(t)
```

```
## [1] 2.441538
```

```
pvalue = 2 * pt(abs(t), n - p - 1, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.01536693
```

```
aux = lm(respiratory_rate ~ pulse + packed_cell_volume + total_protein,
         horse)
fitted = aux$fitted.values
residuals = aux$residuals
tperm = numeric(nperm)
for (k in 1:nperm) {
  Yperm = fitted + sample(residuals)
  perm = lm(Yperm ~ temperature + pulse + packed_cell_volume + total_protein,
            horse)
  tperm[k] = summary(perm)$coefficients["temperature", 3]
}
pvalue = (1 + sum(abs(tperm) > abs(t)))/(1 + nperm)
print(pvalue)
```

```
## [1] 0.01669833
```

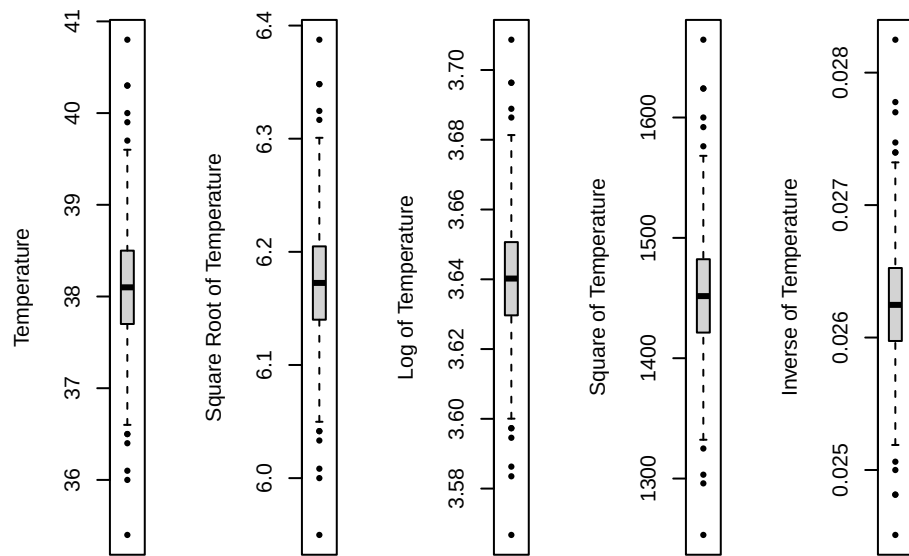
The estimated t test p-value through the permutation method is equal to 0.017. Since the pvalue is less than alpha, we can reject the null hypothesis. The pvalue under the theoretical null distribution is 0.015, which is very similar to the pvalue obtained from the permutation test.

□

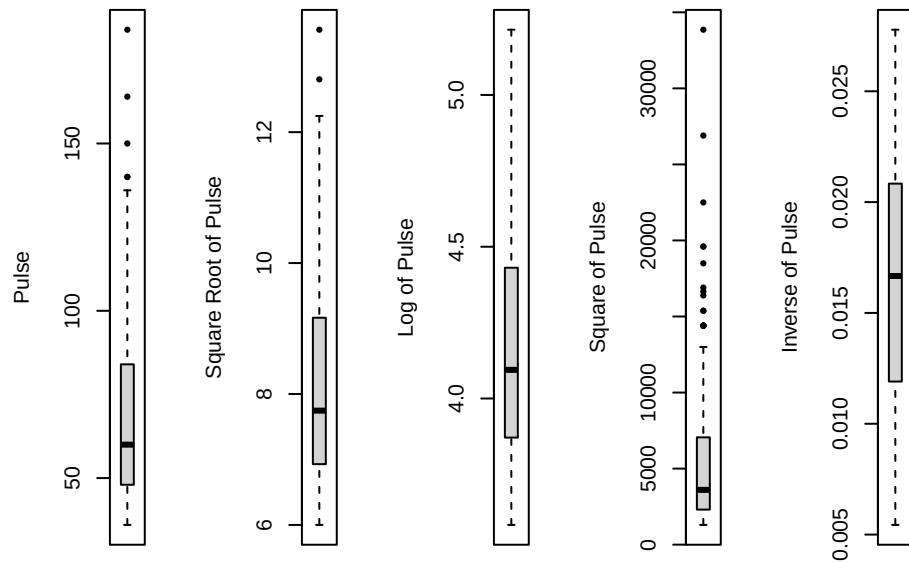
Problem 4. *Proof.*

```
# Check for possible covariate transformations

par(mfrow = c(1, 5))
boxplot(horse$temperature, ylab = "Temperature", pch = 16)
boxplot(sqrt(horse$temperature), ylab = "Square Root of Temperature",
        pch = 16)
boxplot(log(horse$temperature), ylab = "Log of Temperature", pch = 16)
boxplot((horse$temperature)^2, ylab = "Square of Temperature", pch = 16)
boxplot((horse$temperature)^(-1), ylab = "Inverse of Temperature", pch = 16)
```



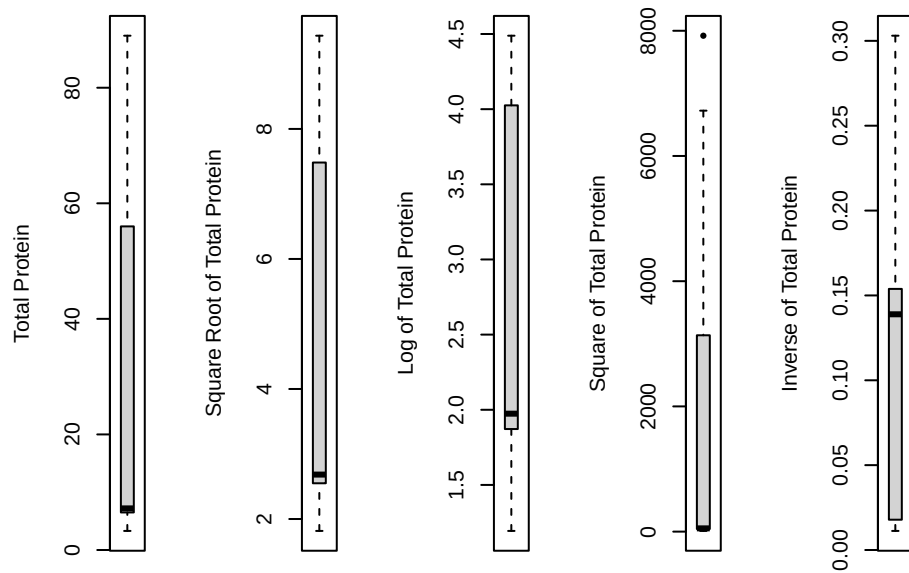
```
boxplot(horse$pulse, ylab = "Pulse", pch = 16)
boxplot(sqrt(horse$pulse), ylab = "Square Root of Pulse", pch = 16)
boxplot(log(horse$pulse), ylab = "Log of Pulse", pch = 16)
boxplot((horse$pulse)^2, ylab = "Square of Pulse", pch = 16)
boxplot((horse$pulse)^(-1), ylab = "Inverse of Pulse", pch = 16)
```



```

boxplot(horse$total_protein, ylab = "Total Protein", pch = 16)
boxplot(sqrt(horse$total_protein), ylab = "Square Root of Total Protein",
  pch = 16)
boxplot(log(horse$total_protein), ylab = "Log of Total Protein", pch = 16)
boxplot((horse$total_protein)^2, ylab = "Square of Total Protein", pch = 16)
boxplot((horse$total_protein)^(-1), ylab = "Inverse of Total Protein",
  pch = 16)

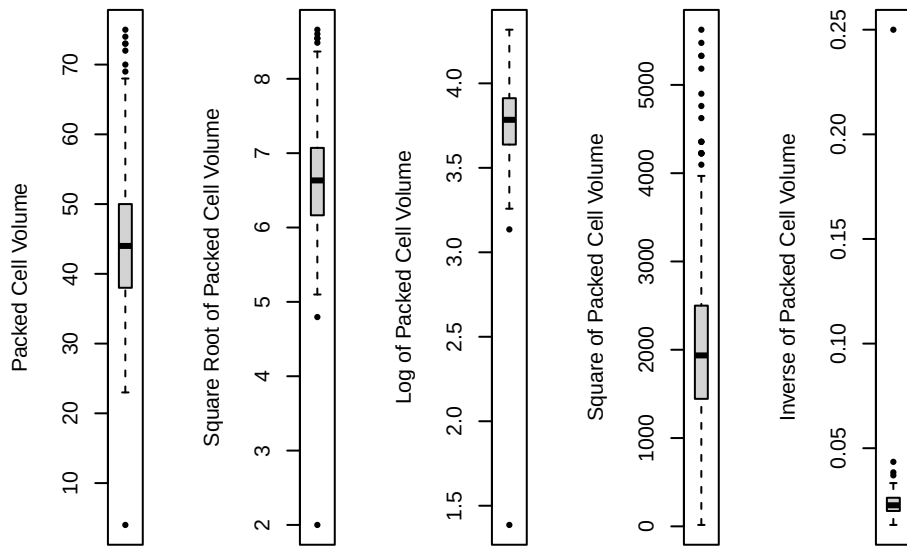
```



```

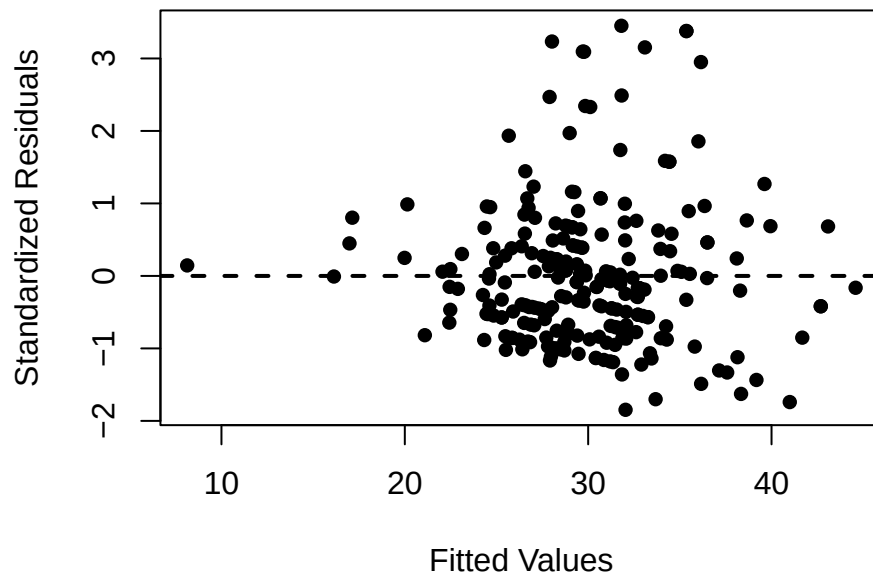
boxplot(horse$packed_cell_volume, ylab = "Packed Cell Volume", pch = 16)
boxplot(sqrt(horse$packed_cell_volume), ylab = "Square Root of Packed Cell Volume",
  pch = 16)
boxplot(log(horse$packed_cell_volume), ylab = "Log of Packed Cell Volume",
  pch = 16)
boxplot((horse$packed_cell_volume)^2, ylab = "Square of Packed Cell Volume",
  pch = 16)
boxplot((horse$packed_cell_volume)^(-1), ylab = "Inverse of Packed Cell Volume",
  pch = 16)

```



Pulse and total protein seem skewed, so need transformations. For pulse, the best transformation is inverse. For total protein, the best transformation is a log transformation. A log transformation is reasonable for packed cell volume to decrease outliers and make for a more symmetric distribution.

```
fit = lm(respiratory_rate ~ temperature + (1/pulse) + log(packed_cell_volume) +
         log(total_protein), horse)
plot(fit$fitted.values, rstandard(fit), xlab = "Fitted Values", ylab = "Standardized Residuals",
     pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



Since constant variance is violated, a transformation should be applied to the response variable.

```
# Box-Cox transformation
horse$inv_pulse = 1/horse$pulse
Y = horse$respiratory_rate

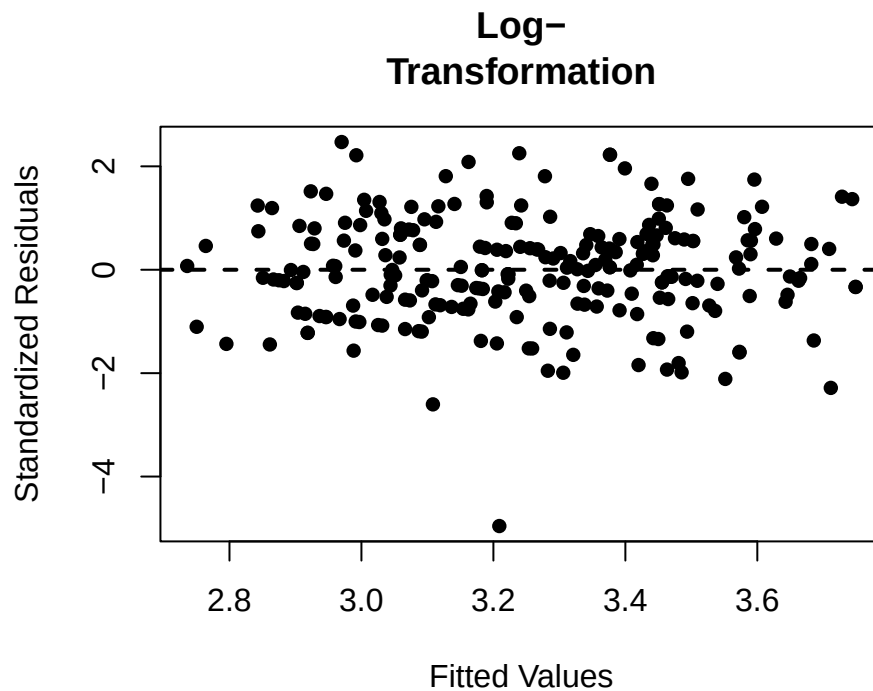
lambda = seq(-2, 2, 0.01)
loglik = numeric(401)
for (i in 1:401) {
  if (lambda[i] == 0) {
    Ypower = log(Y)
  } else {
    Ypower = (Y^lambda[i] - 1)/lambda[i]
  }
  power = lm(Ypower ~ temperature + inv_pulse + log(packed_cell_volume) +
    log(total_protein), horse)
  loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
}

CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] 0.02 0.34
```

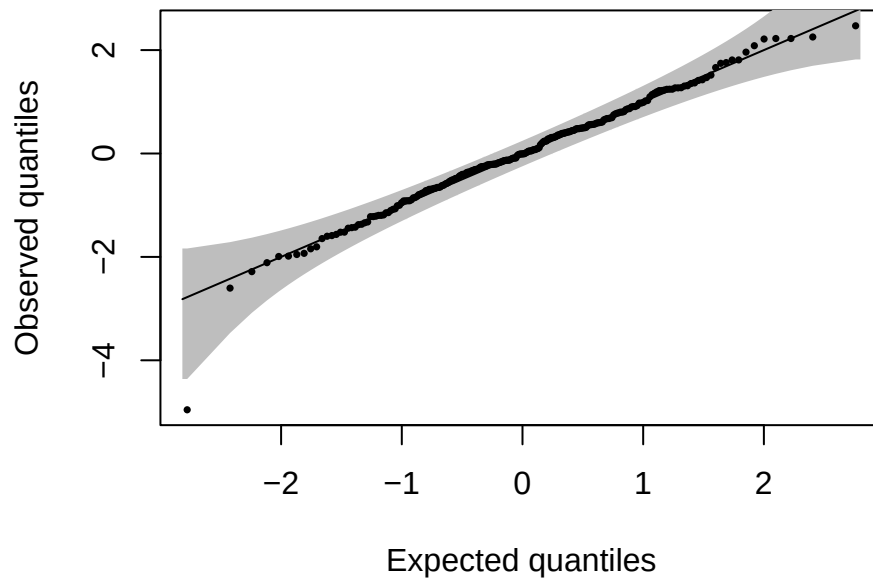

Since the CI is close to covering 0, a log transformation is reasonable.

```
power = lm(log(respiratory_rate) ~ temperature + inv_pulse + log(packed_cell_volume) +  
           log(total_protein), horse)  
plot(power$fitted.values, rstandard(power), main = "Log-  
Transformation",  
      xlab = "Fitted Values", ylab = "  
Standardized Residuals", pch = 16)  
abline(h = 0, lty = 2, lwd = 2)
```



```
library(qqconf)  
qq_conf_plot(rstandard(power), points_params = list(pch = 16, cex = 0.5))
```

```
## no dparams supplied. Estimating parameters from the data...
```



All standardized residuals fall within the band, except one. This is enough to reject normality.

□

Problem 5. *Proof.*

```
# Best subset selection with valid post-selection inference

library(leaps)

# New horse data frame with 5 transformed variables
horse_transformed = data.frame(Y = log(horse$respiratory_rate), temperature = horse$temp,
                               inv_pulse = 1/horse$pulse, log_pcv = log(horse$packed_cell_volume),
                               log_tp = log(horse$total_protein))

Y = horse_transformed$Y
X = horse_transformed[, -1]
n = nrow(X)
P = ncol(X)

train = sample(n, 0.6 * n)
valid = sample(setdiff(1:n, train), 0.2 * n)
test = setdiff(1:n, c(train, valid))
```

```

MSPE = numeric(P)

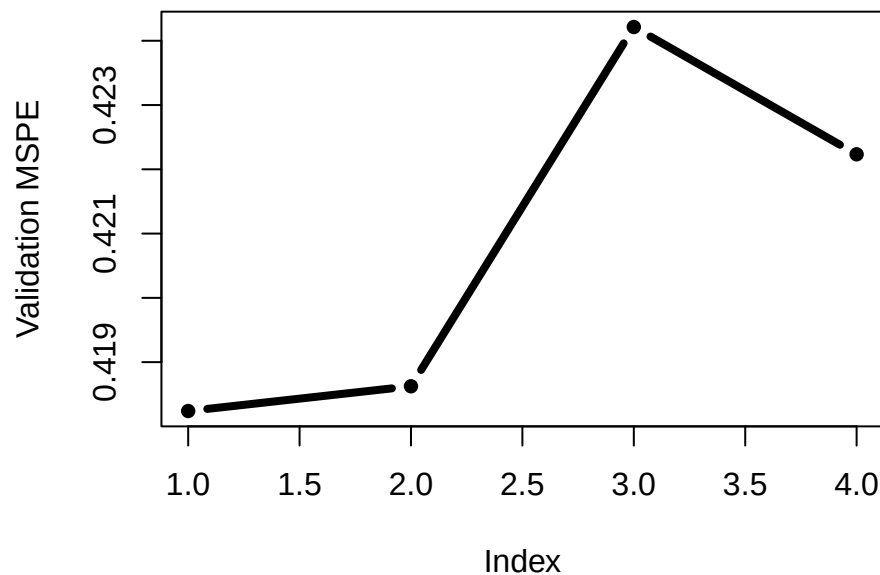
full = lm(Y ~ ., horse_transformed, train)
sub = regsubsets(formula(full), horse_transformed[train, ], nvmax = P)
summary(sub)$which

## (Intercept) temperature inv_pulse log_pcv log_tp
## 1      TRUE      FALSE      TRUE  FALSE  FALSE
## 2      TRUE      TRUE      TRUE  FALSE  FALSE
## 3      TRUE      TRUE      TRUE   TRUE  FALSE
## 4      TRUE      TRUE      TRUE   TRUE   TRUE

for (p in 1:P) {
  fit = lm(Y ~ ., X[, summary(sub)$which[p, -1], drop = FALSE], train)
  preds = predict(fit, X[valid, summary(sub)$which[p, -1], drop = FALSE])
  MSPE[p] = mean((Y[valid] - preds)^2)
}

plot(MSPE, type = "b", pch = 16, lwd = 4, ylab = "Validation MSPE")

```



```

p = which.min(MSPE)
final_model = lm(Y ~ ., X[, summary(sub)$which[p, -1], drop = FALSE],
  test)
summary(final_model)

##
## Call:
## lm(formula = Y ~ ., data = X[, summary(sub)$which[p, -1], drop = FALSE],
##     subset = test)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92884 -0.24592  0.00328  0.24767  1.19785
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.8510      0.2043  18.846  <2e-16 ***
## inv_pulse    -40.0116     12.5067  -3.199   0.0025 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4745 on 46 degrees of freedom
## Multiple R-squared:  0.182, Adjusted R-squared:  0.1642
## F-statistic: 10.23 on 1 and 46 DF, p-value: 0.002497

```

Based on the validation MSPE curve, the full (4 covariates) model provides the most accurate predictor for respiratory rate. Among all of the predictors, inverse pulse (inv_pulse) is the only statistically significant variable on the test set and shows that horses with higher pulse rates tend to have higher respiratory rates. The remaining predictors (temperature, log(packed cell volume), and log(total protein)) are not individually statistically significant.

□

Problem 6. *Proof.*

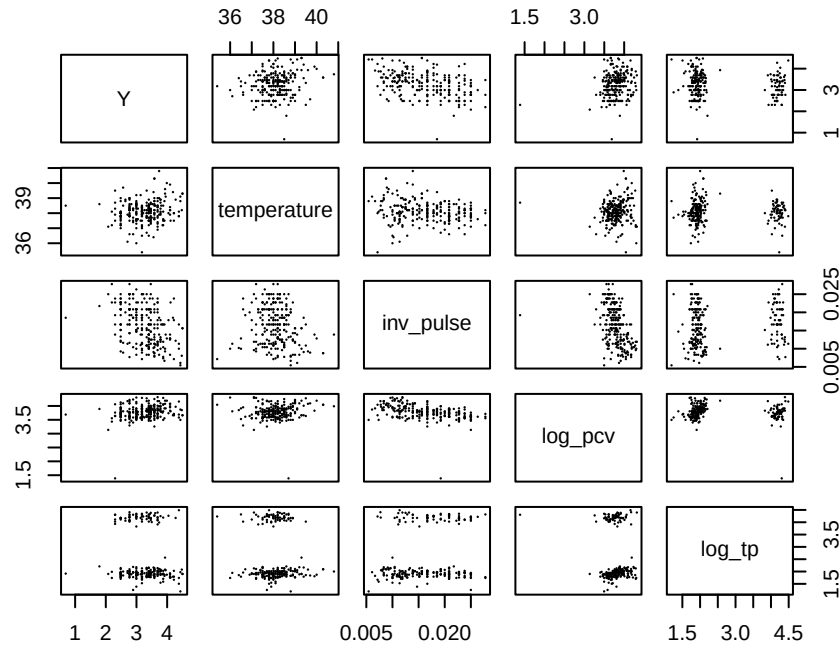
```

# Investigate collinearity
library(faraway)
cor(horse_transformed)

##              Y temperature  inv_pulse   log_pcv    log_tp
## Y              1.0000000   0.16920982 -0.4060674  0.2176732 -0.11911463
## temperature    0.1692098   1.00000000 -0.1751922  0.1032828 -0.04926768
## inv_pulse      -0.4060674  -0.17519222  1.0000000 -0.4159228  0.14147543
## log_pcv         0.2176732   0.10328283 -0.4159228  1.0000000 -0.09311020
## log_tp         -0.1191146  -0.04926768  0.1414754 -0.0931102  1.00000000

```

```
plot(horse_transformed, pch = 16, cex = 0.25)
```



```
cat("VIF:", vif(full))
```

```
## VIF: 1.019564 1.249903 1.235599 1.061112
```

Based on the VIFs and correlation plots, the transformed quantitative variables aren't very collinear.

```
# Ridge Regression
library(glmnet)
train_boot = c(train, valid)
X = as.matrix(X)
cv.ridge = cv.glmnet(X[train_boot, ], Y[train_boot], nfolds = 6, alpha = 0)
ridge = glmnet(X[test, ], Y[test], alpha = 0, lambda = cv.ridge$lambda.min)
print(ridge$beta)
```

```
## 4 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## temperature  0.065253640
## inv_pulse   -26.193003155
## log_pcv      0.135120805
## log_tp      -0.008200548
```

```
# Bootstrap
nboot = 10000
beta = matrix(0, nboot, P)
colnames(beta) = colnames(X)
for (i in 1:nboot) {
  ind = sample(test, replace = TRUE)
  beta[i, ] = glmnet(X[ind, ], Y[ind], alpha = 0, lambda = cv.ridge$lambda.min)$beta[, ]
}
SE = apply(beta, 2, sd)
cat("SE after Bootstrap:", SE, "\n")
```

```
## SE after Bootstrap: 0.06478434 9.057431 0.2336487 0.05209877
```

```
ols = lm(Y ~ ., horse_transformed[train, ])
cat("Summary of OLS coefficients for comparison:", "\n")
```

```
## Summary of OLS coefficients for comparison:
```

```
summary(ols)$coefficients
```

```
##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) -0.21799383 2.11256553 -0.1031891 9.179616e-01
## temperature  0.09222970 0.05287914  1.7441603 8.334173e-02
## inv_pulse   -35.71734705 8.46762134 -4.2181087 4.417755e-05
## log_pcv      0.17708872 0.14653704  1.2084912 2.289102e-01
## log_tp       -0.04042411 0.03929397 -1.0287612 3.053787e-01
```

The ridge regression coefficients all shrunk closer to zero in comparison to the OLS coefficients. The bootstrap results SEs are slightly higher than the SEs calculated using OLS.

□

Problem 7. *Proof.*

```
# Lasso Regression
cv.lasso = cv.glmnet(X[train, ], Y[train], alpha = 1)
lasso = glmnet(X[test, ], Y[test], nfolds = 6, alpha = 1, lambda = cv.lasso$lambda.min)
print(lasso$beta)
```

```
## 4 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## temperature  0.06868708
## inv_pulse   -34.98917710
## log_pcv      0.02454834
## log_tp       .
```

```

for (i in 1:nboot) {
  ind = sample(test, replace = TRUE)
  beta[i, ] = glmnet(X[ind, ], Y[ind], alpha = 1, lambda = cv.lasso$lambda.min)$beta[,
    ]
}
SE = apply(beta, 2, sd)
cat("SE after Bootstrap:", SE, "\n")

```

```
## SE after Bootstrap: 0.08517503 13.86536 0.2958224 0.06885285
```

While some of the estimated lasso coefficients become extremely close to zero, others stay the same. Because the quantitative variables aren't collinear, the estimated SEs of the lasso coefficients are greater than the SEs estimated by the OLS. $\square$

Problem 8. *Proof.*

```

# Investigate the interaction effects between pain and all
# available quantitative covariates on the response variable.

Y = log(horse$respiratory_rate)
inv_pulse = 1/horse$pulse
log_pcv = log(horse$packed_cell_volume)
log_tp = log(horse$total_protein)
horse$pain = as.factor(horse$pain)

fit_interact = lm(Y ~ pain * (temperature + inv_pulse + log_pcv + log_tp),
  horse)
summary(fit_interact)

```

```
##
## Call:
## lm(formula = Y ~ pain * (temperature + inv_pulse + log_pcv +
##   log_tp), data = horse)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-2.50249	-0.28336	0.01694	0.27810	1.23938

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.67381	6.90931	1.834	0.06799 .
paincontinuous	-9.49538	9.31427	-1.019	0.30914

```

## paindepressed      -15.39074      7.84791     -1.961     0.05116 .
## painmild           -10.06434      8.07729     -1.246     0.21412
## painsevere         -11.71858      8.61230     -1.361     0.17504
## temperature        -0.24711      0.17557     -1.407     0.16074
## inv_pulse          -65.69124     24.99086     -2.629     0.00919 **
## log_pcv             0.26388      0.22354      1.180     0.23913
## log_tp             0.04322      0.07810      0.553     0.58059
## paincontinuous:temperature 0.31823      0.24001      1.326     0.18628
## paindepressed:temperature 0.43712      0.19525      2.239     0.02619 *
## painmild:temperature 0.27878      0.20451      1.363     0.17426
## painsevere:temperature 0.30862      0.21187      1.457     0.14668
## paincontinuous:inv_pulse 37.49708     34.78424      1.078     0.28225
## paindepressed:inv_pulse 29.04296     30.93301      0.939     0.34884
## painmild:inv_pulse  28.94192     28.81725      1.004     0.31635
## painsevere:inv_pulse 42.31399     32.33061      1.309     0.19200
## paincontinuous:log_pcv -0.79029      0.55014     -1.437     0.15231
## paindepressed:log_pcv -0.43113      0.45572     -0.946     0.34519
## painmild:log_pcv    -0.22175      0.41562     -0.534     0.59421
## painsevere:log_pcv  -0.13123      0.48412     -0.271     0.78661
## paincontinuous:log_tp -0.04874      0.14272     -0.341     0.73307
## paindepressed:log_tp -0.05449      0.10184     -0.535     0.59316
## painmild:log_tp     -0.09959      0.10090     -0.987     0.32478
## painsevere:log_tp   -0.09975      0.12483     -0.799     0.42509
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5191 on 215 degrees of freedom
## Multiple R-squared:  0.2226, Adjusted R-squared:  0.1358
## F-statistic: 2.565 on 24 and 215 DF,  p-value: 0.0001773

```

```
anova(fit_interact)
```

```

## Analysis of Variance Table
##
## Response: Y
##
##           Df Sum Sq Mean Sq F value    Pr(>F)
## pain         4   5.449   1.3622   5.0559 0.0006506 ***
## temperature   1   1.381   1.3812   5.1262 0.0245650 *
## inv_pulse     1   6.688   6.6882  24.8230 1.29e-06 ***
## log_pcv       1   0.113   0.1131   0.4198 0.5177107
## log_tp       1   0.278   0.2776   1.0305 0.3111925
## pain:temperature 4   1.199   0.2997   1.1122 0.3516623
## pain:inv_pulse  4   0.547   0.1367   0.5075 0.7302813
## pain:log_pcv    4   0.620   0.1549   0.5750 0.6810488

```



```
## pain:log_tp          4  0.310  0.0776  0.2880 0.8855690
## Residuals           215 57.928  0.2694
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
fit_interact = update(fit_interact, . ~ . - pain:log_tp)
summary(fit_interact)
```

```
##
## Call:
## lm(formula = Y ~ pain + temperature + inv_pulse + log_pcv + log_tp +
##      pain:temperature + pain:inv_pulse + pain:log_pcv, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.47892 -0.30174  0.00539  0.29005  1.26364
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    11.50518     6.74017   1.707  0.0892 .
## paincontinuous    -8.04942     8.91973  -0.902  0.3678
## paindepressed   -14.09286     7.68349  -1.834  0.0680 .
## painmild        -8.46462     7.86148  -1.077  0.2828
## painsevere     -11.14258     8.37749  -1.330  0.1849
## temperature     -0.20907     0.16922  -1.235  0.2180
## inv_pulse      -63.75155    24.73401  -2.577  0.0106 *
## log_pcv         0.22270     0.21731   1.025  0.3066
## log_tp         -0.01920     0.03472  -0.553  0.5809
## paincontinuous:temperature  0.27303     0.22872   1.194  0.2339
## paindepressed:temperature  0.39638     0.18971   2.089  0.0378 *
## painmild:temperature   0.23064     0.19719   1.170  0.2434
## painsevere:temperature  0.28201     0.20543   1.373  0.1712
## paincontinuous:inv_pulse 36.91093    32.49695   1.136  0.2573
## paindepressed:inv_pulse  26.90593    30.65544   0.878  0.3811
## painmild:inv_pulse    25.26810    28.37859   0.890  0.3742
## painsevere:inv_pulse   37.27394    31.05793   1.200  0.2314
## paincontinuous:log_pcv -0.74627     0.54443  -1.371  0.1719
## paindepressed:log_pcv  -0.39044     0.45030  -0.867  0.3868
## painmild:log_pcv     -0.20908     0.40982  -0.510  0.6104
## painsevere:log_pcv    -0.06148     0.47249  -0.130  0.8966
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5157 on 219 degrees of freedom
```

```
## Multiple R-squared:  0.2184, Adjusted R-squared:  0.147
## F-statistic:  3.06 on 20 and 219 DF,  p-value: 2.787e-05
```

```
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
pain	4	5.449	1.3622	5.1226	0.0005785 ***
temperature	1	1.381	1.3812	5.1937	0.0236322 *
inv_pulse	1	6.688	6.6882	25.1501	1.095e-06 ***
log_pcv	1	0.113	0.1131	0.4254	0.5149551
log_tp	1	0.278	0.2776	1.0440	0.3080126
pain:temperature	4	1.199	0.2997	1.1269	0.3446763
pain:inv_pulse	4	0.547	0.1367	0.5142	0.7253855
pain:log_pcv	4	0.620	0.1549	0.5826	0.6755637
Residuals	219	58.239	0.2659		

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
fit_interact = update(fit_interact, . ~ . - pain:log_pcv)
summary(fit_interact)
```

```
##
## Call:
## lm(formula = Y ~ pain + temperature + inv_pulse + log_pcv + log_tp +
##      pain:temperature + pain:inv_pulse, data = horse)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-2.48002	-0.30460	0.02382	0.28102	1.25778

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.96315	6.56529	1.974	0.0496 *
paincontinuous	-10.63579	8.68330	-1.225	0.2219
paindepressed	-16.33035	7.30571	-2.235	0.0264 *
painmild	-10.09787	7.55044	-1.337	0.1825
painsevere	-11.96200	7.77452	-1.539	0.1253
temperature	-0.23096	0.16721	-1.381	0.1686
inv_pulse	-63.93125	24.64022	-2.595	0.0101 *
log_pcv	0.05587	0.14724	0.379	0.7047

```
## log_tp -0.02330 0.03429 -0.680 0.4974
## paincontinuous:temperature 0.26405 0.22630 1.167 0.2445
## paindepressed:temperature 0.41385 0.18801 2.201 0.0288 *
## painmild:temperature 0.25287 0.19537 1.294 0.1969
## painsevere:temperature 0.29820 0.20209 1.476 0.1415
## paincontinuous:inv_pulse 41.49455 32.18751 1.289 0.1987
## paindepressed:inv_pulse 33.59660 28.55029 1.177 0.2405
## painmild:inv_pulse 26.27104 27.79406 0.945 0.3456
## painsevere:inv_pulse 36.62324 30.60687 1.197 0.2327
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5137 on 223 degrees of freedom
## Multiple R-squared: 0.2101, Adjusted R-squared: 0.1534
## F-statistic: 3.707 on 16 and 223 DF, p-value: 4.809e-06
```

```
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##          Df Sum Sq Mean Sq F value    Pr(>F)
## pain      4  5.449   1.3622   5.1612 0.000539 ***
## temperature 1  1.381   1.3812   5.2329 0.023101 *
## inv_pulse  1  6.688   6.6882  25.3398 9.9e-07 ***
## log_pcv    1  0.113   0.1131   0.4286 0.513363
## log_tp     1  0.278   0.2776   1.0519 0.306179
## pain:temperature 4  1.199   0.2997   1.1354 0.340652
## pain:inv_pulse 4  0.547   0.1367   0.5180 0.722546
## Residuals 223 58.858   0.2639
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
fit_interact = update(fit_interact, . ~ . - log_pcv)
summary(fit_interact)
```

```
##
## Call:
## lm(formula = Y ~ pain + temperature + inv_pulse + log_tp + pain:temperature +
##      pain:inv_pulse, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -2.48268 -0.31286 0.03051 0.27490 1.26188
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      13.46668      6.41749   2.098 0.03699 *
## paincontinuous   -11.00650      8.61167  -1.278 0.20254
## paindepressed    -16.61034      7.25445  -2.290 0.02297 *
## painmild         -10.42597      7.48643  -1.393 0.16511
## painsevere       -12.14325      7.74500  -1.568 0.11832
## temperature      -0.23872      0.16563  -1.441 0.15091
## inv_pulse        -64.01121     24.59219  -2.603 0.00986 **
## log_tp           -0.02407      0.03416  -0.704 0.48188
## paincontinuous:temperature 0.27420      0.22429   1.223 0.22279
## paindepressed:temperature 0.42211      0.18639   2.265 0.02449 *
## painmild:temperature 0.26196      0.19352   1.354 0.17721
## painsevere:temperature 0.30334      0.20125   1.507 0.13315
## paincontinuous:inv_pulse 41.26294     32.12017   1.285 0.20024
## paindepressed:inv_pulse 32.00850     28.18787   1.136 0.25736
## painmild:inv_pulse 25.47470     27.66172   0.921 0.35807
## painsevere:inv_pulse 36.13291     30.52110   1.184 0.23772
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5128 on 224 degrees of freedom
## Multiple R-squared:  0.2096, Adjusted R-squared:  0.1567
## F-statistic: 3.96 on 15 and 224 DF, p-value: 2.457e-06
```

```
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##              Df Sum Sq Mean Sq F value    Pr(>F)
## pain           4  5.449   1.3622   5.1810 0.0005207 ***
## temperature    1  1.381   1.3812   5.2530 0.0228380 *
## inv_pulse      1  6.688   6.6882  25.4370 9.434e-07 ***
## log_tp         1  0.296   0.2957   1.1247 0.2900581
## pain:temperature 4  1.270   0.3176   1.2078 0.3083174
## pain:inv_pulse  4  0.532   0.1331   0.5062 0.7312140
## Residuals     224 58.896   0.2629
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
fit_interact = update(fit_interact, . ~ . - log_tp)
summary(fit_interact)
```

```
##
## Call:
## lm(formula = Y ~ pain + temperature + inv_pulse + pain:temperature +
##     pain:inv_pulse, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.46671 -0.29546  0.01604  0.28348  1.27608
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      14.0658      6.3538   2.214  0.02785 *
## paincontinuous    -12.1040      8.4601  -1.431  0.15390
## paindepressed    -17.5964      7.1101  -2.475  0.01407 *
## painmild         -10.8063      7.4586  -1.449  0.14878
## painsevere       -13.0211      7.6355  -1.705  0.08951 .
## temperature      -0.2557      0.1637  -1.563  0.11957
## inv_pulse        -64.7889     24.5399  -2.640  0.00887 **
## paincontinuous:temperature  0.3036      0.2201   1.379  0.16925
## paindepressed:temperature  0.4473      0.1827   2.448  0.01512 *
## painmild:temperature   0.2721      0.1928   1.411  0.15948
## painsevere:temperature  0.3269      0.1982   1.649  0.10047
## paincontinuous:inv_pulse 39.6936     32.0069   1.240  0.21621
## paindepressed:inv_pulse 33.3412     28.0928   1.187  0.23655
## painmild:inv_pulse    25.4279     27.6307   0.920  0.35841
## painsevere:inv_pulse   34.7012     30.4192   1.141  0.25518
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5122 on 225 degrees of freedom
## Multiple R-squared:  0.2078, Adjusted R-squared:  0.1585
## F-statistic: 4.216 on 14 and 225 DF,  p-value: 1.399e-06
```

```
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##              Df Sum Sq Mean Sq F value    Pr(>F)
## pain           4  5.449   1.3622   5.1926 0.0005098 ***
```

```
## temperature      1  1.381  1.3812  5.2648 0.0226833 *
## inv_pulse        1  6.688  6.6882 25.4941 9.159e-07 ***
## pain:temperature  4  1.457  0.3643  1.3886 0.2387494
## pain:inv_pulse    4  0.511  0.1277  0.4867 0.7454757
## Residuals        225 59.027  0.2623
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Final Model Equation:

$$\begin{aligned}
Y_i = & \beta_0 + \beta_1 \text{pain}_{i,\text{continuous}} + \beta_2 \text{pain}_{i,\text{depressed}} + \beta_3 \text{pain}_{i,\text{mild}} + \beta_4 \text{pain}_{i,\text{severe}} \\
& + \beta_5 \text{temperature}_i + \beta_6 \text{inv_pulse}_i \\
& + \gamma_1 \text{pain}_{i,\text{continuous}} \cdot \text{temperature}_i + \gamma_2 \text{pain}_{i,\text{depressed}} \cdot \text{temperature}_i + \gamma_3 \text{pain}_{i,\text{mild}} \cdot \text{temperature}_i + \gamma_4 \text{pain}_{i,\text{severe}} \cdot \text{temperature}_i \\
& + \delta_1 \text{pain}_{i,\text{continuous}} \cdot \text{inv_pulse}_i + \delta_2 \text{pain}_{i,\text{depressed}} \cdot \text{inv_pulse}_i + \delta_3 \text{pain}_{i,\text{mild}} \cdot \text{inv_pulse}_i + \delta_4 \text{pain}_{i,\text{severe}} \cdot \text{inv_pulse}_i
\end{aligned}$$

Interpretation of each covariate in the final model:

- $\text{pain}_{\text{continuous}} = -12.1040$ Holding other predictors constant, a horse classified as continuous pain is expected to have a log respiratory rate about 12.10 units lower than a horse with no pain.
- $\text{pain}_{\text{depressed}} = -17.5964$ Holding other predictors constant, a depressed horse has a log respiratory rate about 17.60 units lower than a horse with no pain.
- $\text{pain}_{\text{mild}} = -10.8063$ Holding other predictors constant, a horse with mild pain has a log respiratory rate about 10.81 units lower than a no-pain horse.
- $\text{pain}_{\text{severe}} = -13.0211$ Holding other predictors constant, a horse with severe pain has a log respiratory rate about 13.02 units lower than a no-pain horse.

$\text{-temperature} = -0.2557$ For a horse in the no-pain category with a fixed pulse, a 1°C increase in temperature is associated with a 0.256-unit decrease in log respiratory rate. Temperature is significant overall (ANOVA $p = 0.0227$), but its slope for no-pain horses is not individually significant.

- $\text{inv_pulse} = -64.7889$ Because $\text{inv_pulse} = 1/\text{pulse}$, a decrease in inv_pulse corresponds to a higher pulse rate. For a horse in the no-pain category, holding temperature constant, a one-unit increase in inv_pulse is associated with a 64.79-unit decrease in log respiratory rate.
- $\text{pain}_{\text{depressed:temperature}} = 0.4473$ For depressed horses, the effect of temperature on log respiratory rate is 0.447 units more positive per degree than for no-pain horses. This is the only significant interaction in the model ($p = 0.0151$).

□

Problem 9. *Proof.*

```
# Mixed-effects Model
```

```
Y = horse$respiratory_rate
horse$pain = factor(horse$pain)
horse$surgical_lesion = factor(horse$surgical_lesion)
horse$peristalsis = factor(horse$peristalsis)
horse$abdominal_distension = factor(horse$abdominal_distension)
horse$peripheral_pulse = factor(horse$peripheral_pulse)
horse$temperature_of_extremities = factor(horse$temperature_of_extremities)
horse$surgery <- factor(horse$surgery)
horse$outcome <- factor(horse$outcome)
horse$age <- factor(horse$age)
```

```
# chosen random effects
```

```
# Used transformed variables from problem 8 for reference
```

```
library(lme4)
mixed_eff_mod = lmer(Y ~ temperature + inv_pulse + log_pcv + log_tp +
  outcome + surgery + abdominal_distension + pain + (1 | temperature_of_extremities) +
  (1 | peripheral_pulse) + (1 | age) + (1 | surgical_lesion) + (1 |
  peristalsis), data = horse)
```

```
sigma = data.frame(summary(mixed_eff_mod)$varcor)$sdcor
names(sigma) = data.frame(summary(mixed_eff_mod)$varcor)$grp
```

```
ICC = sigma^2/sum(sigma^2)
print(ICC)
```

```
##                peristalsis                peripheral_pulse
##                0.003159587                0.159798514
## temperature_of_extremities                surgical_lesion
##                0.014265005                0.014474496
##                        age                Residual
##                0.435440514                0.372861885
```

```
summary(mixed_eff_mod, correlation = FALSE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Y ~ temperature + inv_pulse + log_pcv + log_tp + outcome + surgery +
```

```

##      abdominal_distension + pain + (1 | temperature_of_extremities) +
##      (1 | peripheral_pulse) + (1 | age) + (1 | surgical_lesion) +
##      (1 | peristalsis)
##      Data: horse
##
## REML criterion at convergence: 1893.4
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -1.9999 -0.6687 -0.1350  0.5532  3.1234
##
## Random effects:
##      Groups              Name      Variance Std.Dev.
##  peristalsis             (Intercept)    1.706   1.306
##  peripheral_pulse        (Intercept)  86.274   9.288
##  temperature_of_extremities (Intercept)   7.702   2.775
##  surgical_lesion         (Intercept)   7.815   2.795
##  age                     (Intercept) 235.092  15.333
##  Residual                      201.306  14.188
## Number of obs: 240, groups:
## peristalsis, 4; peripheral_pulse, 4; temperature_of_extremities, 4; surgical_lesion,
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)   -78.5493    59.6465  -1.317
## temperature     2.7916     1.4465   1.930
## inv_pulse    -255.3853   265.0700  -0.963
## log_pcv        3.3211     4.2159   0.788
## log_tp        -1.3678     0.9921  -1.379
## outcomeeuthanized  3.3997     3.9336   0.864
## outcomelived     3.5648     2.9265   1.218
## surgery1        2.8395     2.4341   1.167
## abdominal_distensionnone  1.6184     2.7801   0.582
## abdominal_distensionsevere  5.1821     3.6876   1.405
## abdominal_distensionslight  2.3686     2.8540   0.830
## paincontinuous   4.3930     4.3918   1.000
## paindepressed    7.5573     3.4205   2.209
## painmild         2.7623     3.0152   0.916
## painsevere       1.3234     3.9828   0.332

```

```
anova(mixed_eff_mod)
```

```

## Analysis of Variance Table
##              npar  Sum Sq Mean Sq F value

```



```
## temperature      1 2123.98 2123.98 10.5510
## inv_pulse        1  674.08  674.08  3.3485
## log_pcv          1  188.31  188.31  0.9355
## log_tp           1  133.83  133.83  0.6648
## outcome          2  482.84  241.42  1.1993
## surgery          1  270.52  270.52  1.3438
## abdominal_distension 3  546.77  182.26  0.9054
## pain             4 1271.03  317.76  1.5785
```

```
# Box Cox
```

```
Y = horse$respiratory_rate
```

```
lambda = seq(-2, 2, 0.01)
```

```
loglik = numeric(401)
```

```
for (i in 1:401) {
```

```
  if (lambda[i] == 0) {
```

```
    Ypower = log(Y)
```

```
  } else {
```

```
    Ypower = (Y^lambda[i] - 1)/lambda[i]
```

```
  }
```

```
  power = lmer(Ypower ~ temperature + inv_pulse + log_pcv + log_tp +
```

```
    outcome + surgery + abdominal_distension + pain + (1 | temperature_of_extremities)
```

```
    (1 | peripheral_pulse) + (1 | age) + (1 | surgical_lesion) +
```

```
    (1 | peristalsis), horse)
```

```
  loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
```

```
}
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## unable to evaluate scaled gradient
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge: degenerate Hessian with 1 negative eigenvalues
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## unable to evaluate scaled gradient
```

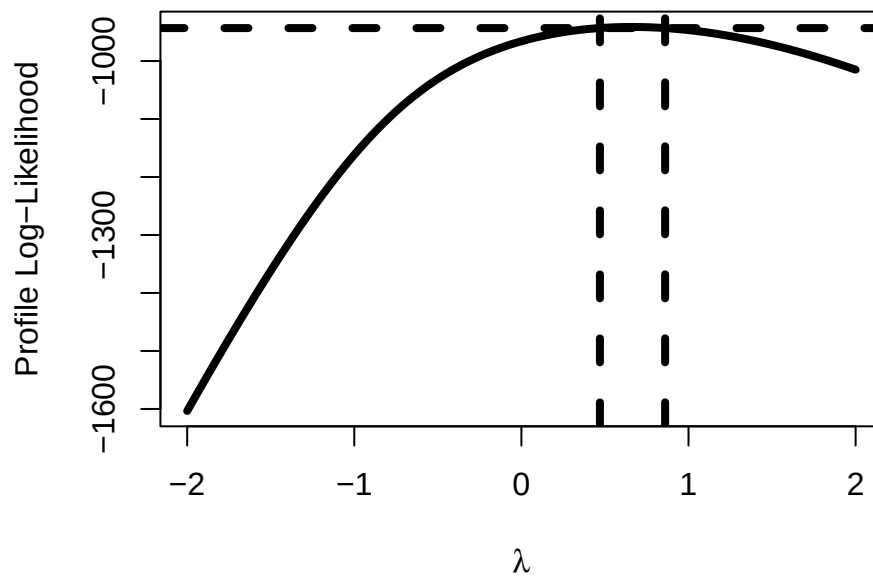
```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge: degenerate Hessian with 1 negative eigenvalues
```

```
alpha = 0.05
```

```
CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] 0.47 0.86
```

```
plot(lambda, loglik, "l", xlab = expression(lambda), ylab = "
Profile Log-Likelihood",
     lwd = 4)
abline(h = max(loglik) - qchisq(0.95, 1)/2, lty = 2, lwd = 4)
abline(v = CI, lty = 2, lwd = 4)
```



The covariates chosen as random effects are age, peripheral_pulse, peristalsis, and temperature_of_extremities, since these were the categorical variables with the largest nonzero variance components in the mixed-effects model. All categorical variables with exactly zero variance components were treated as fixed effects, as instructed.

A suitable transformation for our response variable is.

The total variation in your response variable explained by each random effect is as follows: abdominal_distension (variance = 0.02668, 2.67%) and temperature_of_extremities (variance = 0.00652, 0.65%), and surgery (variance = 0.005579, 0.56%).

Another interpretation for each ICC: ICC(abdominal_distension) = 0.0936 Two horses with the same abdominal distension level have an expected correlation of about 9.4% in their log respiratory rates. This points too a moderate grouping effect of adominal distension level on respiratory rate.

$ICC(\text{temperature_of_extremities}) = 0.0229$ Two horses with the same extremity temperature category have an expected correlation of about 2.3% in their log respiratory rates. This points to a very small grouping effect of extremity temperature on respiratory rate.

$ICC(\text{surgery}) = 0.01956768$ Two horses that underwent the same type of surgery have an expected correlation of about 2.0% in their log respiratory rates. This points to a very small grouping effect of surgery on respiratory rate.

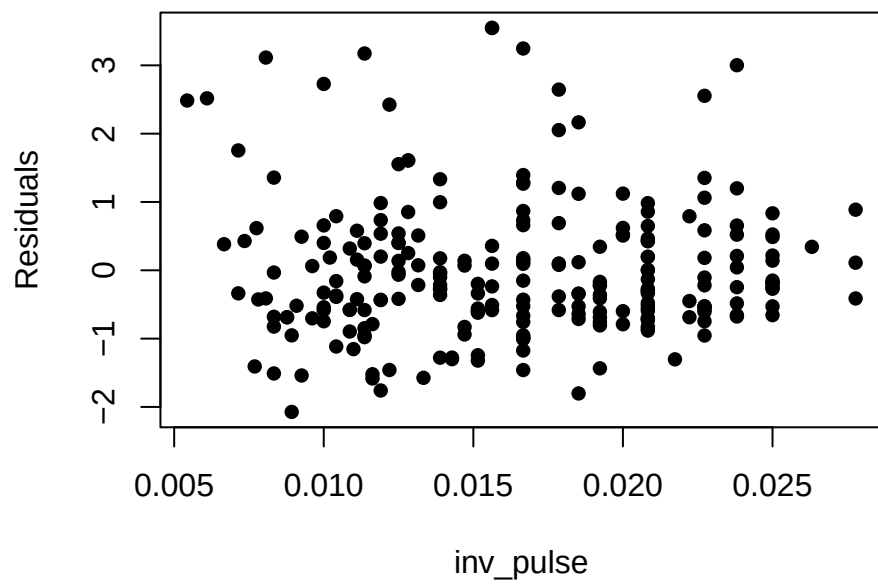
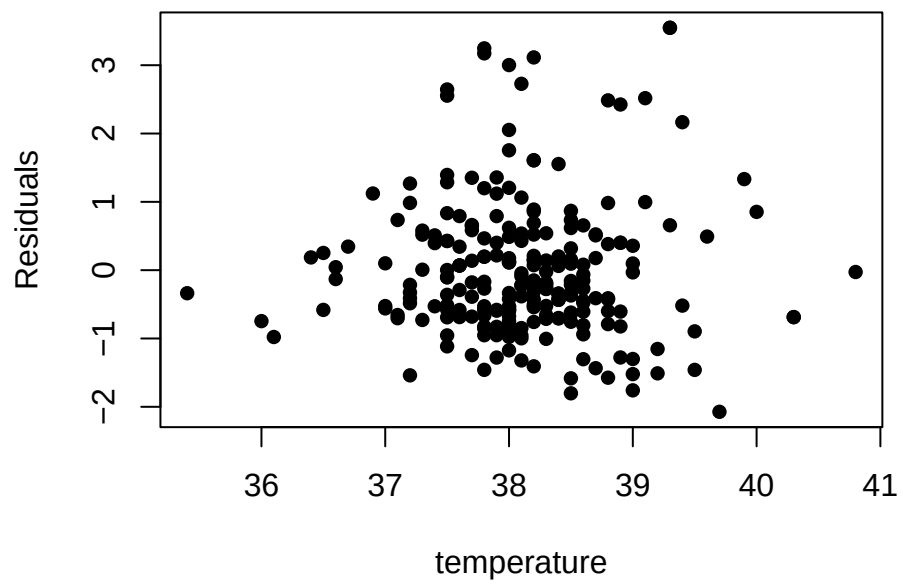
□

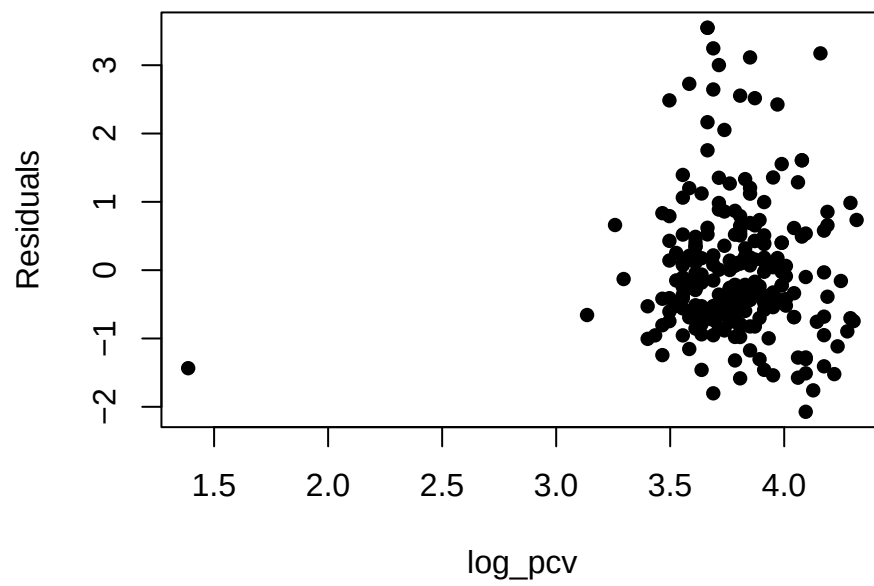
Problem 10. *Proof.*

```
base_model <- lm(Y ~ temperature + inv_pulse + log_pcv + log_tp, data = horse)
print(summary(base_model))
```

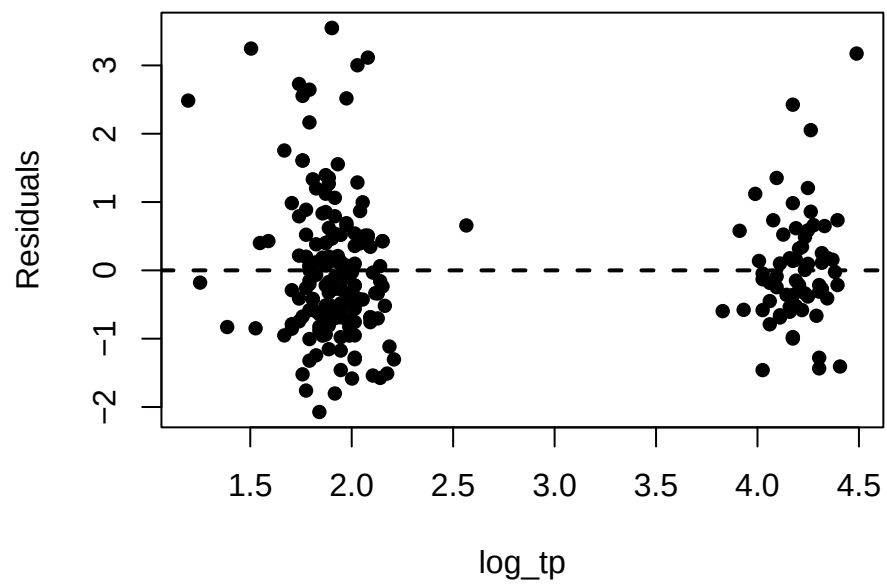
```
##
## Call:
## lm(formula = Y ~ temperature + inv_pulse + log_pcv + log_tp,
##     data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.539  -9.389  -3.283   6.711  54.318
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -73.7142    56.4478  -1.306   0.1929
## temperature     3.3214     1.4041   2.365   0.0188 *
## inv_pulse    -1126.6554   211.5286  -5.326 2.34e-07 ***
## log_pcv        -0.2385     4.2153  -0.057   0.9549
## log_tp         -1.3970     0.9729  -1.436   0.1523
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.45 on 235 degrees of freedom
## Multiple R-squared:  0.1778, Adjusted R-squared:  0.1638
## F-statistic: 12.71 on 4 and 235 DF,  p-value: 2.238e-09
```

```
Y = log(horse$respiratory_rate)
X = model.matrix(base_model)
n = dim(X)[1]
p = dim(X)[2] - 1
residuals = base_model$residuals
plot(rstandard(base_model) ~ temperature + inv_pulse + log_pcv + log_tp,
     horse, ylab = "Standardized
Residuals", pch = 16)
```





```
abline(h = 0, lty = 2, lwd = 2)
```



```
# Studentized Breusch-Pagan Test
Yaux = residuals(base_model)^2
aux = lm(Yaux ~ temperature + inv_pulse + log_pcv + log_tp, horse)
BP = n * summary(aux)$r.squared
print(BP)
```

```
## [1] 11.83406
```

```
pvalue = pchisq(BP, p, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.01862882
```

```
library(lmtest)
bptest(base_model)
```

```
##
## studentized Breusch-Pagan test
##
## data: base_model
## BP = 11.834, df = 4, p-value = 0.01863
```

```
# Brown-Forsythe Test -- pain
```

```
library(car)
leveneTest(Y ~ pain, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  4  5.0709 0.0006161 ***
##      235
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Brown-Forsythe Test -- outcome
```

```
library(car)
leveneTest(Y ~ outcome, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  2  1.2748 0.2814
##      237
```

```
# Brown-Forsythe Test -- temp of extremities
```

```
library(car)
leveneTest(Y ~ temperature_of_extremities, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    3  0.9573 0.4136
##           236
```

```
# Brown-Forsythe Test -- peripheral pulse
```

```
library(car)
leveneTest(Y ~ peripheral_pulse, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    3  2.7338 0.0444 *
##           236
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Brown-Forsythe Test -- abdominal distension
```

```
library(car)
leveneTest(Y ~ abdominal_distension, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    3  1.7893 0.1499
##           236
```

```
# Brown-Forsythe Test -- peristalsis
```

```
library(car)
leveneTest(Y ~ peristalsis, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    3  2.4855 0.06136 .
##           236
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For the quantitative variables, a studentized Breusch–Pagan test was conducted. Because the p-value is well above 0.05, we fail to reject the null hypothesis of constant variance. So here is no evidence of heteroscedasticity, and the residual variance does not appear to depend on any of the included quantitative predictors.

For the qualitative variables, a Brown–Forsythe tests was conducted.

The Brown–Forsythe test yields $p = 0.0006161$, providing statistically significant evidence that the residual variance differs across pain groups (heteroscedasticity).

The Brown–Forsythe test yields $p = 0.2814$, so there is no statistically significant evidence of differing variances across outcome groups (homoscedasticity).

The Brown–Forsythe test yields $p = 0.4136$, so there is no statistically significant evidence that the residual variance differs across peripheral-pulse categories.

The Brown–Forsythe test yields $p = 0.0444$, indicating statistically significant evidence that residual variances differ across abdominal-distension groups (heteroscedasticity).

The Brown–Forsythe test yields $p = 0.061$, so there is no statistically significant evidence of variance differences across peristalsis groups at the 5% level. The Brown–Forsythe test yields $p = 0.061$, so there is no statistically significant evidence that the residual variance differs across peristalsis groups at the 5% level.

□

Problem 11. *Proof.*

```
Y = log(horse$respiratory_rate)
quant_model = lm(Y ~ temperature + inv_pulse + log_pcv + log_tp, horse)

ols_summary = summary(quant_model)
```

```
X = model.matrix(quant_model)
n = nrow(X)
p = ncol(X)
beta = coef(quant_model)
residuals = quant_model$residuals
```

```
vcovOLS = summary(quant_model)$sigma^2 * summary(quant_model)$cov.unscaled
vcovOLS
```

```
##           (Intercept)  temperature  inv_pulse      log_pcv      log_tp
## (Intercept)  3.465801376 -8.178581e-02 -3.95918767 -0.0717941594 -4.198632e-03
## temperature -0.081785813  2.144548e-03  0.04609586 -0.0002127801  3.544896e-05
## inv_pulse   -3.959187671  4.609586e-02  48.66851858  0.3874615003 -2.437183e-02
## log_pcv     -0.071794159 -2.127801e-04  0.38746150  0.0193275056  1.661329e-04
## log_tp      -0.004198632  3.544896e-05 -0.02437183  0.0001661329  1.029491e-03
```



```
confint(quant_model)
```

```
##              2.5 %      97.5 %  
## (Intercept) -3.02626049  4.30911449  
## temperature -0.01560159  0.16686712  
## inv_pulse   -51.63124263 -24.14315363  
## log_pcv     -0.16066009  0.38712271  
## log_tp      -0.09471723  0.03170743
```

```
# Heteroscedasticity-Consistent Estimators
```

```
library(sandwich)  
vcovHC3 = vcovHC(quant_model, type = "HC3")  
vcovHC3
```

```
##              (Intercept)  temperature  inv_pulse      log_pcv      log_tp  
## (Intercept)  3.362777937 -7.536590e-02 -3.74147070 -1.139449e-01 -2.075313e-03  
## temperature -0.075365899  2.105212e-03  0.01147321 -1.268062e-03  4.728520e-06  
## inv_pulse   -3.741470703  1.147321e-02  50.04492660  6.903247e-01 -4.612666e-02  
## log_pcv     -0.113944921 -1.268062e-03  0.69032469  3.977325e-02  1.043878e-05  
## log_tp      -0.002075313  4.728520e-06 -0.04612666  1.043878e-05  9.672646e-04
```

```
alpha = 0.05  
HC3_CI = cbind(beta - qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)), beta +  
               qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)))  
  
colnames(HC3_CI) = colnames(confint(quant_model))  
HC3_CI
```

```
##              2.5 %      97.5 %  
## (Intercept) -2.97133693  4.25419093  
## temperature -0.01476098  0.16602651  
## inv_pulse   -51.82423720 -23.95015906  
## log_pcv     -0.27967231  0.50613493  
## log_tp      -0.09277705  0.02976725
```

```
library(lmtest)  
confint(coeftest(quant_model, vcov. = vcovHC3))
```

```
##              2.5 %      97.5 %  
## (Intercept) -2.97133693  4.25419093
```

```
## temperature -0.01476098 0.16602651
## inv_pulse -51.82423720 -23.95015906
## log_pcv -0.27967231 0.50613493
## log_tp -0.09277705 0.02976725
```

```
test = matrix(0, p, 4)
rownames(test) = names(beta)
colnames(test) = c("Estimate", "Std. Error", "t value", "Pr(>|t|)")

test[, 1] = beta
test[, 2] = sqrt(diag(vcovHC3))
test[, 3] = test[, 1]/test[, 2]
test[, 4] = 2 * pt(abs(test[, 3]), n - p, lower.tail = FALSE)

test
```

```
##           Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  0.64142700 1.83378787  0.3497826 7.268152e-01
## temperature  0.07563277 0.04588259  1.6483980 1.006073e-01
## inv_pulse   -37.88719813 7.07424389 -5.3556534 2.028092e-07
## log_pcv      0.11323131 0.19943232  0.5677681 5.707345e-01
## log_tp      -0.03150490 0.03110088 -1.0129908 3.121064e-01
```

```
hc3_test = coeftest(quant_model, vcov. = vcovHC3)
```

```
results_table <- data.frame(Coefficient = rownames(ols_summary$coefficients),
  Estimate = ols_summary$coefficients[, "Estimate"], OLS_SE = ols_summary$coefficients[,
    "Std. Error"], HC3_SE = sqrt(diag(vcovHC3)), t_HC3 = hc3_test[,
    "t value"], p_HC3 = hc3_test[, "Pr(>|t|)"])

results_table
```

```
##           Coefficient      Estimate    OLS_SE    HC3_SE    t_HC3
## (Intercept) (Intercept)  0.64142700 1.86166629 1.83378787  0.3497826
## temperature temperature  0.07563277 0.04630927 0.04588259  1.6483980
## inv_pulse   inv_pulse -37.88719813 6.97628258 7.07424389 -5.3556534
## log_pcv     log_pcv  0.11323131 0.13902340 0.19943232  0.5677681
## log_tp     log_tp -0.03150490 0.03208568 0.03110088 -1.0129908
##           p_HC3
## (Intercept) 7.268152e-01
## temperature 1.006073e-01
## inv_pulse   2.028092e-07
## log_pcv     5.707345e-01
## log_tp     3.121064e-01
```

The HC3 standard errors are larger for most of the coefficients. (Intercept) OLS SE: 1.86166629 HC3 SE: 1.83378787 temperature OLS SE:0.04630927 HC3 SE:0.04588259 pulse OLS SE: 6.97628258 HC3 SE:7.07424389 PCV OLS SE: 0.13902340 HC3 SE: 0.13902340 total_protein OLS SE: 0.03208568 HC3 SE:0.03208568

As the SEs increased, the HC3 t-values got smaller. (Intercept): -2.145 to -1.965 temperature: 2.442 to 2.205 pulse: 5.967 to 5.144 PCV: -0.640 to -0.568 total_protein: -1.011 to -0.972.

The p-values got larger. (Intercept) 0.0330 to 0.0506 (no longer < 0.05) temperature 0.0154 to 0.0284 (still < 0.05) pulse $8.82e-09$ to $5.67e-07$ (still extremely significant)

PCV 0.5228 to 0.5709 (still NS) total_protein 0.3132 to 0.3319 (still NS)

The intercept is the only coefficient whose significance status changed. In the OLS (Intercept) is significant ($p = 0.033$), but HC3 shows it is only marginally significant ($p = 0.0506$).

The HC3 CIs are consistently wider:

(Intercept) OLS [-216.74, -9.22]; HC3 [-226.25, 0.29]. Which now covers zero. temperature: OLS [0.656, 6.142]; HC3 [0.362, 6.43] pulse: OLS [0.169, 0.336]; HC3 [0.156, 0.349] PCV: OLS [-0.283, 0.144]; HC3 [-0.310, 0.171] total_protein: OLS [-0.109, 0.0353]; HC3 [-0.112, 0.038]

The qualitative conclusions remained the same for temperature, pulse, PCV and total_protein.

□

Problem 12. *Proof.*

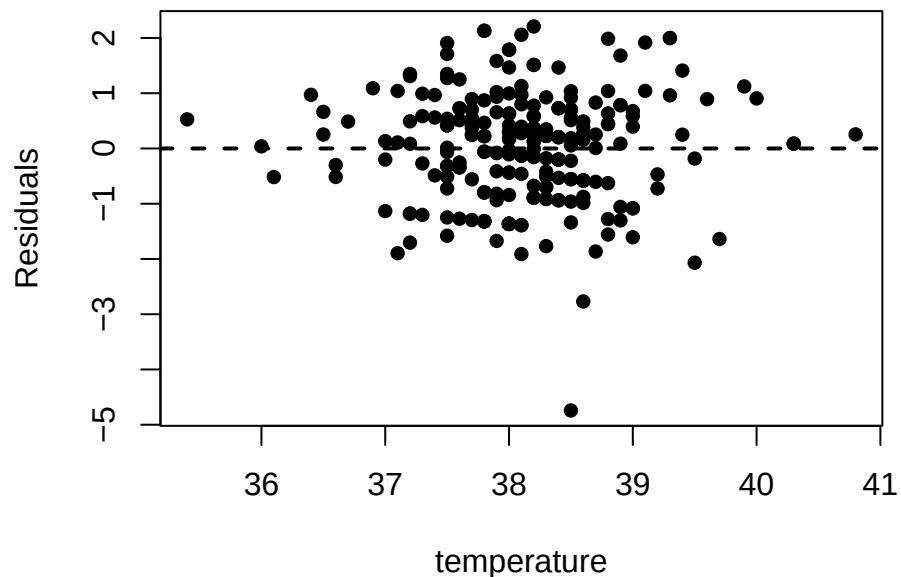
```
# Weighted LS using temperature
fit = lm(Y ~ temperature, horse)
summary(fit)
```

```
##
## Call:
## lm(formula = Y ~ temperature, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.60814 -0.38480  0.07008  0.35146  1.21521
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.72473     1.87907  -0.918  0.35962
## temperature   0.13055     0.04929   2.649  0.00862 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5515 on 238 degrees of freedom
## Multiple R-squared:  0.02863,    Adjusted R-squared:  0.02455
## F-statistic: 7.015 on 1 and 238 DF,  p-value: 0.008623
```

```

X = model.matrix(fit)
n = dim(X)[1]
p = dim(X)[2] - 1
residuals = fit$residuals
plot(rstandard(fit) ~ temperature, horse, ylab = "Standardized
Residuals",
     pch = 16)
abline(h = 0, lty = 2, lwd = 2)

```



The variance of the residuals appears to be an increasing function of temperature. Since the weights are inversely proportional to the residual variances, we infer that the weights would need to be a decreasing function of temperature.

```

w = rep(1, n)
err = Inf
tol = 1e-10
niter = 0
maxiter = 1000
while (err > tol & niter < maxiter) {
  aux = lm(log(residuals^2) ~ log(temperature), horse)
  err = sum(abs(w - exp(-aux$fitted.values)))/sum(w)
  w = exp(-aux$fitted.values)
}

```

```

WLS = lm(Y ~ temperature, horse, weights = w)
residuals = WLS$residuals
niter = niter + 1
}
summary(aux)

```

```

##
## Call:
## lm(formula = log(residuals^2) ~ log(temperature), data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.6695  -0.9516   0.3997   1.4790   4.2089
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -36.344     25.195  -1.443   0.150
## log(temperature)   9.326      6.921   1.348   0.179
##
## Residual standard error: 2.032 on 238 degrees of freedom
## Multiple R-squared:  0.007572,    Adjusted R-squared:  0.003402
## F-statistic: 1.816 on 1 and 238 DF,  p-value: 0.1791

```

```
summary(WLS)
```

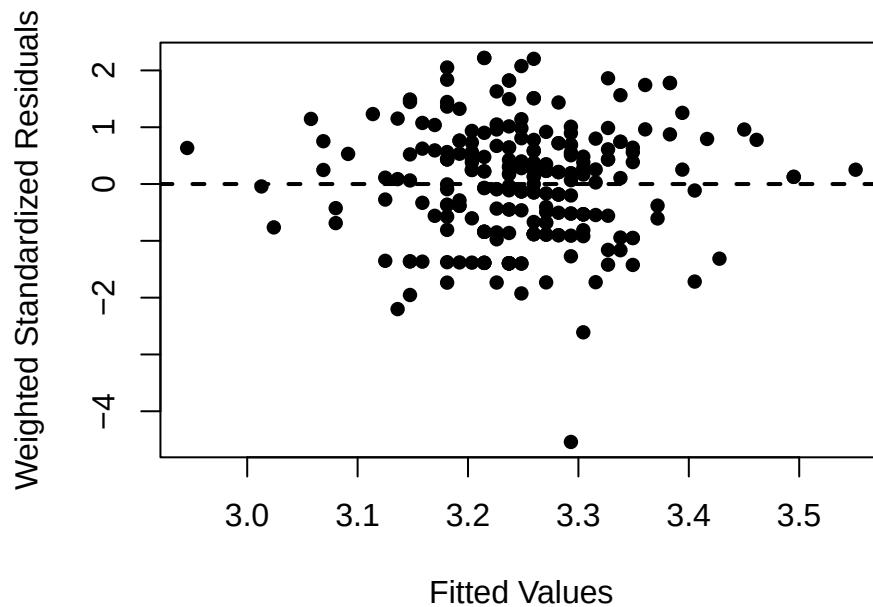
```

##
## Call:
## lm(formula = Y ~ temperature, data = horse, weights = w)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
##  -8.2025  -1.2296   0.2316   1.1614   4.0115
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.02349     1.82516  -0.561   0.5755
## temperature   0.11212     0.04804   2.334   0.0204 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.812 on 238 degrees of freedom
## Multiple R-squared:  0.02238,    Adjusted R-squared:  0.01827
## F-statistic: 5.448 on 1 and 238 DF,  p-value: 0.02042

```

We estimate that $\sigma_i^2 \approx 1.64 \times 10^{-16} \cdot \text{temperature}_i^{9.33}$

```
plot(WLS$fitted.values, rstandard(WLS), xlab = "Fitted Values", ylab = "Weighted Standard  
      pch = 16)  
abline(h = 0, lty = 2, lwd = 2)
```



```
WLS_all = lm(Y ~ temperature + inv_pulse + log_pcv + log_tp, horse, weights = w)  
summary(WLS_all)
```

```
##  
## Call:  
## lm(formula = Y ~ temperature + inv_pulse + log_pcv + log_tp,  
##     data = horse, weights = w)  
##  
## Weighted Residuals:  
##      Min       1Q   Median       3Q      Max   
## -7.952 -1.024 -0.029   1.025   4.173   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   0.10632    1.79804   0.059   0.9529      
## temperature    0.08797    0.04452   1.976   0.0493 *  
```

```
## inv_pulse    -36.80128    6.79445   -5.416   1.5e-07 ***
## log_pcv      0.12386     0.13978    0.886    0.3765
## log_tp      -0.02853     0.03124   -0.913    0.3621
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.673 on 235 degrees of freedom
## Multiple R-squared:  0.1768, Adjusted R-squared:  0.1628
## F-statistic: 12.62 on 4 and 235 DF,  p-value: 2.579e-09
```

The WLS standard errors are slightly smaller than the corresponding OLS estimate.

```
# t-test and CIs
```

```
t_WLS = coef(WLS_all)/summary(WLS_all)$coefficients[, "Std. Error"]
t_WLS
```

```
## (Intercept) temperature  inv_pulse    log_pcv    log_tp
##  0.05913009  1.97605238 -5.41637149  0.88611701 -0.91323043
```

```
CI_WLS = confint(WLS_all, level = 0.95)
CI_WLS
```

```
##                2.5 %        97.5 %
## (Intercept) -3.436017e+00  3.64865376
## temperature  2.645365e-04  0.17567304
## inv_pulse    -5.018710e+01 -23.41546165
## log_pcv      -1.515217e-01  0.39924636
## log_tp       -9.007966e-02  0.03301842
```

```
t_OLS = coef(quant_model)/summary(quant_model)$coefficients[, "Std. Error"]
t_OLS
```

```
## (Intercept) temperature  inv_pulse    log_pcv    log_tp
##  0.3445446   1.6332101  -5.4308577   0.8144766  -0.9818990
```

```
CI_OLS = confint(quant_model, level = 0.95)
CI_OLS
```

```
##                2.5 %        97.5 %
## (Intercept) -3.02626049  4.30911449
## temperature -0.01560159  0.16686712
## inv_pulse    -51.63124263 -24.14315363
## log_pcv      -0.16066009  0.38712271
## log_tp       -0.09471723  0.03170743
```

Temperature became significant under WLS, where it was insignificant in OLS. Inv_pulse is highly significant in both. The other coefficients don't change much in significance. The temperature CI under WLS does not include 0, and it does in OLS. WLS gives slightly narrower CIs for all covariates compared to OLS.

□

Problem 13. *Proof.*

```
# mean comparisons for pain
Q = factor(horse$pain)
n = length(Q)
K = length(levels(Q))
mu = aggregate(Y ~ pain, data = horse, FUN = function(x) mean(x))[, 2]
sigma = sqrt(sum((Y - mu[Q])^2)/(n - K))
SE = sigma * sqrt(2 * K/n)
diff = outer(mu, mu, "-")
diff = diff[lower.tri(diff)]
lab = outer(levels(Q), levels(Q), function(x, y) {
  paste0(x, "-", y)
})
names(diff) = lab[lower.tri(lab)]
CI = cbind(diff - qt(1 - alpha/2, n - K) * SE, diff + qt(1 - alpha/2,
  n - K) * SE)
colnames(CI) = c("Lower Bound", "Upper Bound")
print(CI)
```

```
##              Lower Bound  Upper Bound
## continuous-alert      0.23287316  0.668893975
## depressed-alert       0.16919893  0.605219750
## mild-alert            -0.03972276  0.396298053
## severe-alert          0.07428923  0.510310049
## depressed-continuous -0.28168463  0.154336183
## mild-continuous       -0.49060633 -0.054585513
## severe-continuous     -0.37659433  0.059426483
## mild-depressed        -0.42693211  0.009088712
## severe-depressed      -0.31292011  0.123100708
## severe-mild           -0.10399841  0.332022404
```

```
# Tukey's Honestly Significant Difference Method cannot do manual
# calculations because group sizes are unequal
TukeyHSD(aov(Y ~ pain, horse))
```

```
## Tukey multiple comparisons of means
```



```
##      95% family-wise confidence level
##
## Fit: aov(formula = Y ~ pain, data = horse)
##
## $pain
##              diff          lwr          upr          p adj
## continuous-alert  0.45088357  0.09060383  0.81116330  0.0061313
## depressed-alert    0.38720934  0.09141242  0.68300626  0.0035506
## mild-alert         0.17828764 -0.10471823  0.46129352  0.4164565
## severe-alert       0.29229964 -0.05089540  0.63549468  0.1356510
## depressed-continuous -0.06367423 -0.40476160  0.27741315  0.9859950
## mild-continuous   -0.27259592 -0.60265213  0.05746029  0.1582325
## severe-continuous  -0.15858393 -0.54150377  0.22433592  0.7859018
## mild-depressed     -0.20892170 -0.46705234  0.04920895  0.1741052
## severe-depressed   -0.09490970 -0.41789877  0.22807937  0.9280458
## severe-mild        0.11401200 -0.19730528  0.42532927  0.8520905
```

Tukey's Honestly Significant Difference Method (Tukey's HSD) calculates pairwise differences using a critical value from Tukey's studentized range distribution rather than the t-distribution. It can only take positive values and is parametrized by the number of groups (k) and the residual ($n-k$) degrees of freedom. For each pair of group means $\hat{\mu}_k - \hat{\mu}_\ell$, simultaneous 95% CIs are computed such that $\text{FWER} \leq \alpha$ using the upper α -quantile of the studentized range divided by $\sqrt{2}$ and multiplied by the pooled sample standard deviation times the square root of 2 times the number of groups divided by the sample size, eg $\hat{\mu}_k - \hat{\mu}_\ell \pm \frac{R_{K,n-K;\alpha}}{\sqrt{2}} \cdot S\sqrt{\frac{2K}{n}}$.

The CIs using the t-distribution are generally wider than the narrower Tukey's estimates. Tukey's estimates take into account the studentized range distribution, which adjusts for the fact that multiple comparisons are being made simultaneously while controlling the FWER. Since Tukey's is more conservative, there are less significant comparisons under this method. Only 2 comparisons are significant under Tukey's (continuous-alert and depressed-alert). These CIs do not contain 0. For the CIs using the t-distribution, both of those comparisons are significant, as well as two others (severe-alert and mild-continuous). $\square$

Problem 14. *Proof.*

$\square$